

INITIAL STAGE OF WATER IMPACT

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INTRODUCTION

The problem of interaction between a solid body and a liquid with a free surface is a broad subject that includes several significant sections of classical and modern hydrodynamics. In this review we confine our discussion to processes characterized by strong unsteadiness in their development and by the existence of a contact line between the free surface of the liquid and the solid-body surface.

Detailed investigation of such processes can be applied to problems in ship building, aviation, and energetics (e.g. slamming problems, and the problems of the landing of flying boats and of high-velocity liquid-drop impact upon structural elements). In a general form, such processes can be described in the following manner: At the initial instant of time $t = 0$, a solid body touches a free surface of liquid. At this moment the position of the solid body, the domain occupied by the liquid, and the velocity field of the liquid particles are assumed to be known. For $t > 0$, either the law of body motion or the external forces affecting it are prescribed. The liquid flow and the character of its action upon the body are to be determined.

Interest in this field of hydrodynamics arose more than a half century ago in connection with the problem of the landing of flying boats. The first theories of solid-body impact with a liquid (the penetration theory of von Kármán and the impact theory of Sedov) were directed at a global description of this process. Many applied problems have since been solved on the basis of these theories. But in some cases, more complete information about the process is required. Thus, in studying material erosion it is important to know the stress distribution in the contact zone between the liquid and the solid-body surface. It is also necessary to take into account the peculiarities of the flow velocity field in order to determine the

height and form of a free-surface splash. It should be noted that very often the global characteristics can be determined with good accuracy from only rather simplified knowledge of the interaction mechanism, and therefore a detailed description of the process in some problems is unnecessary.

The problem of a blunt body penetrating a liquid that initially occupies a lower half-space and is at rest is a typical example of the type considered here. In fact, this very problem is the focus of this review. It should be noted that even with further simplifications (the liquid is assumed to be ideal and incompressible, its flow to be potential, and the solid body to be rigid), the problem is still very complicated for the following reasons:

1. the flow region is not predetermined;
2. the division of its boundary into components ("wetted part of a solid-body surface" and "free boundary") is also unknown;
3. singularities may appear on the three-phase contact line.

Quantitative information about the process of interaction, even with the use of idealized models, can be obtained only using numerical methods. But to understand the dynamics of the process and develop an adequate computational algorithm, it is necessary to investigate analytically the qualitative picture of the phenomenon, obtaining asymptotic solutions for characteristic stages of the process that are difficult and inexpedient to derive by numerical methods.

The initial stage of interaction, which immediately follows the instant of contact between the liquid and the body, is of particular interest. At this stage the main values are significantly changed, which determines the further evolution of the process. In some cases these changes are of shock character, which permits us to simplify the description. Experimental studies (Moghisi & Squire 1981, Eroshin et al. 1984) show that the loads affecting a blunt body from a liquid are maximum at rather small values of the dimensionless time t . (The ratio of the radius of body curvature at the tangency point to the submergence velocity is taken as the time scale.) Figure 1 shows a typical relation between drag force and time. Thus the initial stage of interaction gives the necessary estimates for the dynamic-strength calculation of a structure and also supplies the initial data for further numerical calculation of the hydrodynamic problem.

The initial stage of impact is also of interest because of changes in the flow topology at $t = 0$: A previously absent component of the liquid boundary adjacent to the solid body appears. The problem studied presents a typical and relatively simple case of such a situation. It should be mentioned that at present there are no mathematically rigorous results of a general character in the theory of nonlinear unsteady hydrodynamic problems with a free boundary with a contact line. Thus the following



Figure 1 Typical relation between drag force and time (Moghisi & Squire 1981).

asymptotic expansions of the solution of the impact problem are of a formal character. The general approach is based on the method of matched asymptotic expansions, which is very useful in both fluid and gas dynamics (Van Dyke 1964). At the same time, problems of impact between solid and liquid masses have their own peculiarities, e.g. usually they do not include an explicit small parameter, its role being played by a small time.

We note that the description of the initial stages of liquid-drop impact upon a solid surface is very similar to that of a body penetrating a liquid. It also resembles the case of contact problems in elasticity theory (Galin 1980). Thus we may combine the drop-impact and body-penetration theories when analyzing the initial stage of impact.

This review consists of three parts. The first deals with the initial asymptotics of the process of body entrance into an ideal, incompressible liquid occupying a half-space, for which the Lagrangian approach is used. The focus here is on the three-dimensional effects and the fine flow structure in the vicinity of the three-phase contact line.

The second part is devoted to the influence of liquid compressibility during the initial-impact stage. The main problem here is to construct an asymptotic solution as the Mach number tends to zero. Only more recent results on the drop-impact problem are discussed, as there are already interesting reviews available on this topic (Lesser & Field 1983, Gonor & Rivkind 1982).

The third part deals with more complicated models of impact. The effects of the elasticity of a submerging structure and of the existence of fixed solid boundaries are discussed. At the end of this review we present estimates showing that the effects of gravity, viscosity, and capillarity are negligible in the initial stage.

ENTRY OF A SOLID BODY INTO AN IDEAL INCOMPRESSIBLE FLUID

Formulation of the Problem

We consider here the initial stage of unsteady motion due to entry of a solid body into a liquid. Initially ($t = 0$) the fluid occupies a lower half-

space and is at rest as the solid body touches its free surface. The liquid is assumed to be ideal, incompressible, and without surface tension; external mass forces are assumed absent. The domain $\Omega(t)$ occupied by the liquid changes with time, and at any instant $t > 0$ its boundary consists of the free surface $[\Sigma_1(t)]$, the wetted part of the penetrating body surface $[\Sigma_2(t)]$, and the contact line between them $[\Gamma(t)]$. Let us suppose that domains Σ_1 and Σ_2 are connected within some time interval $0 < t < T$. This assumption is physically justified and is used in all available models of the penetration process. The liquid motion is assumed to be potential, because under our assumptions the mechanism of vorticity generation is absent.

Using Euler variables, we can formulate the problem for the velocity potential $\varphi(x, y, z, t)$ as follows:

$$\Delta\varphi = 0 \quad \text{in } \Omega(t), \quad (1)$$

$$F_t + \nabla\varphi \cdot \nabla F = 0 \quad \text{on } \Sigma_1(t), \quad (2)$$

$$2\varphi_t + |\nabla\varphi|^2 = 0 \quad \text{on } \Sigma_1(t), \quad (3)$$

$$(\nabla\varphi - \mathbf{V}) \cdot \mathbf{n} = 0 \quad \text{on } \Sigma_2(t), \quad (4)$$

$$\nabla\varphi \rightarrow 0 \quad \text{as } x^2 + y^2 + z^2 \rightarrow \infty, \quad (5)$$

$$\varphi = 0 \quad \text{at } t = 0, \quad (6)$$

$$\Sigma_1 = \{z = 0, x' + y' > 0\} \quad \text{at } t = 0, \quad (7)$$

$$\Sigma_2 = \emptyset \quad \text{at } t = 0. \quad (8)$$

Here x, y, z are Cartesian coordinates, the relation $F(x, y, z, t) = 0$ determines the free boundary Σ_1 when $t > 0$, $\mathbf{n}$ is the external unit normal to the submerged body surface, and $\mathbf{V}$ is the velocity of points on that surface. If the body is rigid and the law of its motion is given, then $\mathbf{n}$ and $\mathbf{V}$ are known functions of the coordinates and time. This very problem is discussed in most theoretical studies of the initial stage of entry. An important special case of condition (4) corresponds to normal penetration of a body into a liquid; in this case, we have $\mathbf{V} = (0, 0, -V)$, where V depends only on t . (The flow pattern is presented in Figure 2.) Conditions (7) and (8) exclude initial contact of a body with liquid on a set having nonzero planar measure, i.e. the case of a flat body. In this case, instead of (7) and (8), we should use

$$\Sigma_1 = \{z = 0, (x, y) \in D\} \quad \text{when } t = 0, \quad (7')$$

$$\Sigma_2 = \{z = 0, (x, y) \in \mathbb{R}^2 \setminus \bar{D}\} \quad \text{when } t = 0, \quad (8')$$

where D is a planar domain having a compact complement. If the function φ is known, then the liquid-velocity vector $\mathbf{v}$ is given by gradient $\nabla\varphi$, and

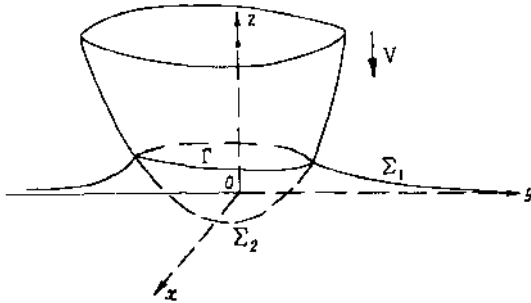


Figure 2 The flow pattern for normal penetration of a body into a liquid.

the pressure p is found from the Cauchy-Lagrange integral

$$p = -\rho(\varphi_t + |\nabla\varphi|^2/2),$$

where ρ is the liquid density.

As was noted, the existence of the unknown boundary of the domain $\Omega(t)$ presents the main difficulty in analyzing problem (1)–(8). It is possible to avoid this by using Lagrangian coordinates ξ, η, ζ in place of the Euler coordinates. [Nalimov & Pukhnachov (1975) have applied Lagrangian coordinates to problems of the dynamics of an ideal liquid with a free boundary.] In Lagrangian variables the unknown functions are the liquid pressure p and the coordinates $\mathbf{x} = (x, y, z)$ of the liquid particle holding a position $\xi = (\xi, \eta, \zeta)$ at $t = 0$. Denote $\Omega(0) = \Omega_0$; according to (7) and (8), Ω_0 coincides with the half-space $\zeta < 0$. Theoretically, there are two possibilities at small t . The first is characterized by the transition of liquid particles from the free boundary to the solid surface with time. In this case the preimage of the wetted part of the body surface $\Sigma_2(t)$ when mapping $\xi \rightarrow \mathbf{x}$ [denote it $\sigma_2(t)$] belongs to the boundary $\zeta = 0$ of the domain Ω_0 , and the preimage $\sigma_1(t)$ of the free surface $\Sigma_1(t)$ is the complement to the plane $\zeta = 0$ of the set $\bar{\sigma}_2(t)$ (see Figure 3). The second possibility requires that the preimage of the set $\Sigma_2(t)$ possess two components: The first, $\sigma_2(t)$ as before, belongs to the plane $\zeta = 0$, and the second, $\sigma_0(t)$, comprises internal points of the domain Ω_0 . Analysis of the solution asymptotics of problem (1)–(8) as $t \rightarrow 0$ shows that the first case seems to apply for blunt bodies (Pukhnachov 1979, Korobkin 1985). Analysis of the classical Wagner problem of wedge entry into water (Hughes 1972, Johnstone & Mackie 1973) suggests that the second case is typical for pointed bodies with constant entry velocity.

We now formulate the entry problem in Lagrangian variables under the assumption that the first flow scheme is valid. For $t \in (0, T)$, we must find

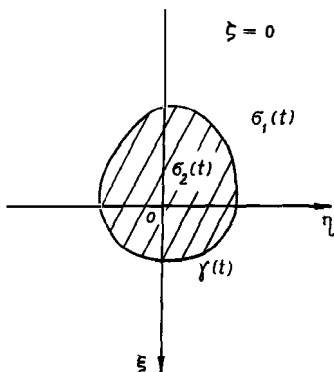


Figure 3 The scheme of the flow-region boundary in Lagrangian coordinates

the curve $\gamma(t)$ dividing the domains $\sigma_1(t)$ and $\sigma_2(t)$ in the plane $\zeta = 0$ and the solution $\mathbf{x}(\xi, t)$, $p(\xi, t)$ of the Euler equations

$$J^* \cdot \mathbf{x}_{tt} + \rho^{-1} \nabla_{\xi} p = 0, \quad \det J = 1, \quad (9)$$

in the domain Ω_0 satisfying the following boundary and initial conditions:

$$p = 0 \quad \text{on} \quad \sigma_1(t), \quad (10)$$

$$(\mathbf{x}_t - \mathbf{V}) \cdot \mathbf{n} = 0 \quad \text{on} \quad \sigma_2(t), \quad (11)$$

$$\mathbf{x} - \xi \rightarrow 0 \quad \text{as} \quad \xi \rightarrow \infty, \quad (12)$$

$$\mathbf{x} = \xi, \mathbf{x}_t = 0 \quad \text{at} \quad t = 0, \quad (13)$$

$$\sigma_1 = \{\zeta = 0, \xi^2 + \eta^2 > 0\}, \sigma_2 = \emptyset \quad \text{at} \quad t = 0. \quad (14)$$

Here $J = \partial(\mathbf{x})/\partial(\xi)$ is the Jacobian matrix of the mapping $\xi \rightarrow \mathbf{x}$, J^* is the matrix conjugate to J , and ∇_{ξ} is the gradient with respect to the variables ξ, η, ζ . As before, $\mathbf{n}$ and $\mathbf{V}$ denote the unit vector normal to the body surface and the velocity of a surface point, respectively. They are considered given functions of x, y, z , and t .

Lagrangian coordinates are most useful for investigating the initial stage of entry of blunt bodies into water under the assumption of no flattening. It is this problem that we turn to next.

The Blunt Body

The methods of analysis applied in penetration theory depend significantly on the geometrical properties of the solid body. When we classify according to these properties, the most general case is that of a blunt body without flattening. For this case the principal curvatures k_1, k_2 of the body surface at the point of initial contact satisfy the condition $0 < k_2 \leq k_1 < \infty$.

It should be noted that until now, all considerations of this problem

have been restricted to the cases of plane and axisymmetric bodies. The basic results for the plane problem were obtained by Wagner (1932). His method was based on reducing the plane analog of problem (1)–(8) to that of impact on a free surface of a floating plate whose width changes in time as $2a(t)$. The unknown plate width can be found from the condition of continuous joining of the free and solid parts of the liquid boundary. It can be shown that this condition is practically equivalent to the finiteness condition of fluid-particle displacement. Wagner's idea was further developed by Schmieden (1953), who considered the axisymmetrical problem of a body of revolution entering an ideal incompressible fluid.

The limits of applicability of the Wagner theory and its possible refinements have been analyzed in detail by Logvinovich (1969). In particular, he describes the transition stage of submergence of a body of finite dimensions, when the Wagner asymptotics is invalid, but the regime of body flow with jet separation is far from steady. One other important question studied by Logvinovich is that of a free-falling body decelerating due to liquid resistance. An additional impact mass $m(t)$ (Shiffman & Spencer 1945a,b) is used in this problem. Theoretical estimates made by Logvinovich (1969) and Lotov (1971) and their comparison with the experimental data obtained by Richardson (1948) demonstrate that the influence of the m vs. t dependence on the dynamics of a submerging blunt body is essential.

The Wagner theory permits predictions to be made of the integral characteristics of the process, such as the drag force or the kinetic energy of the liquid. It is also widely used when calculating the dynamic responses of liquid to submerging elastic shells [See the book by Grigolyuk & Gorshkov (1976), and references therein.] At the same time, the Wagner theory possesses some shortcomings. First, it is not clear how the Wagner approach may be generalized to essentially three-dimensional problems. Besides, the Wagner method leads to an incorrect description of the flow picture in the vicinity of the three-phase contact line $\Gamma(t)$. In particular, it leads to predictions of an unlimited increase in pressure as $\Gamma(t)$ is approached from the wetted part of the body surface.

A new approach to the initial asymptotics of the problem of blunt-body entry has been proposed by Pukhnachov (1979). It is based on the assumption of a small displacement of liquid particles, $\mathbf{x} - \boldsymbol{\xi} = \mathbf{X}$, at small t and enables Equations (9)–(14), written in Lagrangian coordinates near the initial state for definition of the leading asymptotic terms, to be linearized. The boundary-value problem remains nonlinear, since the curve $\gamma(t)$ [the preimage of the contact line $\Gamma(t)$ on the plane $\zeta = 0$] remains unknown. In addition, the appearance of singularities of the solution on the line $\gamma(t)$ is possible.

This approach was developed by Korobkin (1985), who considered the problem of normal blunt-body entry with constant velocity V . He found that the leading term in the asymptotic solution of problem (9)–(14) as $t \rightarrow 0$ is the same as if the submerging body were an elliptic paraboloid:

$$2z = k_1 x^2 + k_2 y^2 - 2Vt.$$

The displacement vector in the linearized problem will be potential: $(X, Y, Z) = \mathbf{X} = \nabla_\xi \phi$. The pressure is connected with the displacement potential ϕ by the relation $p = -\rho \phi_{tt}$. After linearization, conditions (10) and (11) admit integration in t . As a result, we obtain the following boundary-value problem for the function ϕ :

$$\Delta \phi = 0 \quad \text{when} \quad \zeta < 0, \quad (15)$$

$$\phi = 0 \quad \text{on} \quad \sigma_1(t), \quad (16)$$

$$2\phi_\zeta = k_1 \xi^2 + k_2 \eta^2 - 2Vt \quad \text{on} \quad \sigma_2(t), \quad (17)$$

$$\phi \rightarrow 0 \quad \text{as} \quad \xi \rightarrow \infty. \quad (18)$$

Its solution is to be found together with the curve $\gamma(t)$ dividing the domains σ_1 and σ_2 on the plane $\zeta = 0$.

Here time appears as a parameter, which is a remarkable feature of problem (15)–(18). The form of boundary condition (17) means that the curve γ will be found in the class of ellipses with semi-minor axis $b(k_1^{-1} Vt)^{1/2}$ and eccentricity e , where b and e are unknown parameters. To define them, we add to (15)–(18) an additional condition

$$\iiint_{\zeta < 0} |\nabla_\xi \phi_{,i}|^2 d\xi d\eta d\zeta < \infty, \quad (19)$$

which expresses the finiteness of the liquid kinetic energy (Pukhnachov 1979). Problem (15)–(19) allows separation of variables in ellipsoidal coordinates and thus permits us to construct its exact solution in terms of Lamé functions (Korobkin 1985). The expression for the constant $b > 0$ has the form $b^2 = 6(1 - e^2)(2 - e^2 - \varepsilon^2)^{-1}$, where $\varepsilon = (1 - k_2 k_1^{-1})^{1/2}$ is the eccentricity of the section $z = \text{constant}$ that touches the paraboloidal body at its top. The relative positions of the curve γ and the section indicated are shown in Figure 4. The formula

$$e = \varepsilon \sqrt{\frac{2 - \varepsilon^2}{3 - 2\varepsilon^2}}$$

defines the relation between the eccentricities e and ε . Figure 5 presents a plot of the function $e(\varepsilon)$. In the limiting case $k_1 = k_2$, $\varepsilon = 0$ (a body of

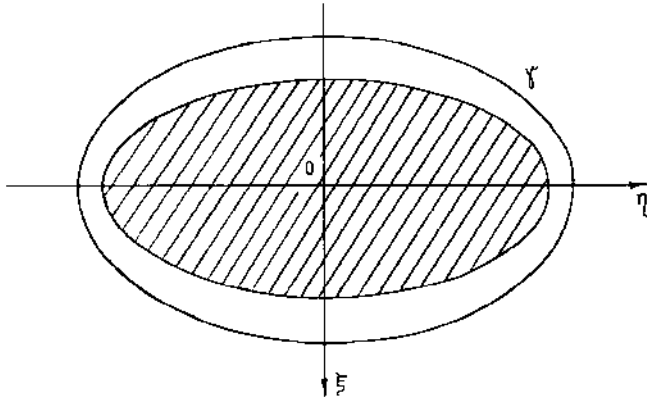


Figure 4 Relative arrangement of the contact line γ and the section $z = 0$ of the elliptic paraboloid touching the body at the apex.

revolution), we obtain $r = (3k_1^{-1}Vt)^{1/2}$ for the contact spot radius, which is in agreement with the results of Schmieden (1953). The other limiting case, $k_2 \rightarrow 0$, $\varepsilon \rightarrow 1$, corresponds to the transition to the plane problem. In this case, the half-width of the contact zone is $a = 2(k_1^{-1}Vt)^{1/2}$, which conforms to the classical Wagner result.

A posteriori analysis shows that the image $\xi \rightarrow \mathbf{x}$ is univalent in the domain $\zeta < 0$ at small $t > 0$, and that the liquid particles do not penetrate

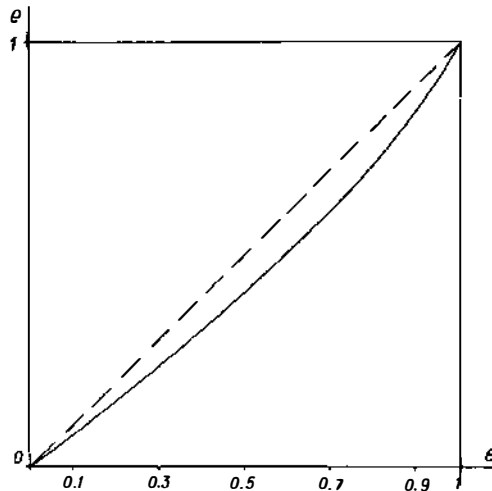


Figure 5 A plot of the function $e = e(\epsilon)$.

into the "forbidden" region occupied by the solid body. The asymptotics constructed satisfies relations (9)–(14), correct to $O(t^{1/2})$ as $t \rightarrow 0$, everywhere except for a narrow domain near the curve $\gamma(t)$. In this domain, the given asymptotics obviously is invalid, since the corresponding velocity and pressure fields have singularities at the points γ . The construction of an inner expansion is necessary for defining more exactly the flow structure near the contact line. The estimates made with the help of the method of matched asymptotic expansions by Pukhnachov & Korobkin (1981) show that the dimensions of the vicinity of the curve γ in which nonlinear terms are to be taken into account are of order $t^{3/2}$ as $t \rightarrow 0$. The boundary-value problem for the leading term of the inner expansion is difficult to solve in the general case. But in the plane case, it can be effectively investigated (Korobkin & Pukhnachov 1985). The results obtained are presented below.

Let us consider dimensionless variables, taking for length, time, and pressure scales the values k^{-1} , kV^{-1} and ρV^2 , respectively (retaining the original notation for the dimensionless values). Here $k > 0$ is the curvature of a plane contour at the point of initial contact. Further, we assume $Y = 0$ and consider the values X , Z , and p independent of η .

In the plane problem the contact set $\gamma(t)$ consists of the two points $\xi = \pm a(t)$, $\zeta = 0$; in the case under consideration, we have $a = 2t^{1/2}$ (Wagner 1932). In the vicinity of the right contact point, we define the inner variables α , β , according to the formulae $\xi = 2t^{1/2} + t^{3/2}\alpha$, $\zeta = t^{3/2}\beta$. The inner expansions of the functions X , Z , and p are sought as follows:

$$\begin{aligned} X &= t^{3/2}Q(\alpha, \beta) + t^2Q^{(1)}(\alpha, \beta) + \dots, \\ Z &= t + t^{3/2}P(\alpha, \beta) + t^2P^{(1)}(\alpha, \beta) + \dots, \\ p &= t^{-1}\Pi(\alpha, \beta) + t^{-1/2}\Pi^{(1)}(\alpha, \beta) + \dots. \end{aligned} \quad (20)$$

The powers of t multiplying the functions P , Q , and Π are determined by matching the leading terms in (20) with the leading terms in the outer expansion, the expressions for which in Lagrangian coordinates were found by Pukhnachov (1979). The function Π is connected with P and Q by the Cauchy–Lagrange integral analog:

$$P_\alpha^2 + Q_\alpha^2 + 2Q_\alpha + 2\Pi = 0. \quad (21)$$

The functions P and Q satisfy the relations

$$P_{\alpha\alpha} - Q_{\alpha\beta} + P_\beta P_{\alpha\alpha} - P_\alpha P_{\alpha\beta} + Q_\beta Q_{\alpha\alpha} - Q_\alpha Q_{\alpha\beta} = 0, \\ P_\beta + Q_\alpha + P_\beta Q_\alpha - P_\alpha Q_\beta = 0 \quad \text{when } \beta < 0, \quad (22)$$

$$P_\alpha^2 + Q_\alpha^2 + 2Q_\alpha = 0 \quad \text{when } \alpha > 0, \beta = 0. \quad (23)$$

$$P_\alpha = 0 \quad \text{when } \alpha < 0, \beta = 0, \quad (24)$$

which follow from (9)–(11) and (21). The matching condition yields the outer limit of the inner expansion as follows:

$$(\alpha + i\beta)^{-1/2}(P + iQ) \rightarrow -2 \quad \text{as} \quad \alpha + i\beta \rightarrow \infty \quad (25)$$

(assuming $\alpha^{1/2} > 0$ when $\alpha > 0$). The one-sided inequality $\zeta + Z - 0.5(\xi + X)^2 + t \leq 0$, which implies that the moving liquid particles cannot enter the “forbidden” region limited by a moving incompressible contour in inner variables, is reduced to

$$P + \beta \leq 0 \quad \text{for} \quad \beta \leq 0. \quad (26)$$

Since the region where the inner expansion is constructed disappears as $t \rightarrow 0$, initial conditions (12) and (13) do not lead to additional limitations on the functions P and Q .

Relations (22)–(26) form a nonlinear boundary-value problem with limitations on the (α, β) plane. The main difficulty in the study of this problem is that the quasi-linear system (22) is of a combined type: For any solution P, Q of this system, there exist two families of complex characteristics and one real family $\beta = \text{constant}$. Nevertheless, problem (22)–(26) allows for a full investigation. This is achieved by transition to the generalized hodograph plane λ, μ , where $\lambda = P_\alpha, \mu = Q_\alpha$ are the new independent variables, and $\alpha = A(\lambda, \mu), \beta = B(\lambda, \mu)$ are assumed to be the new required functions (Korobkin & Pukhnachov 1985). The domain of the solution definition in the hodograph plane is a half-circle $S = \{\lambda, \mu: \lambda < 0, \lambda^2 + \mu^2 + 2\mu < 0\}$. By virtue of (22)–(25), the functions A, B form the solution of the following boundary-value problem:

$$[\lambda^2 + (1 + \mu)^2](A_\lambda B_\mu - A_\mu B_\lambda) = B_\lambda^2 + B_\mu^2,$$

$$B_{\lambda\lambda} + B_{\mu\mu} = 0 \quad \text{at} \quad (\lambda, \mu) \in S,$$

$$B = 0 \quad \text{if} \quad (\lambda, \mu) \in \partial S,$$

$$(\lambda + i\mu)(A + iB)^{1/2} \rightarrow -1 \quad \text{as} \quad \lambda + i\mu \rightarrow 0.$$

The problem obtained admits an exact solution. The function B is calculated explicitly as $4B = -\text{Im}[1 + 2i(\lambda + i\mu)^{-1}]^2$, and the function A is expressed by a quadrature. This solution carries out a univalent image of the domain $S \setminus \{\lambda, \mu: \lambda = 0, -2 < \mu \leq \mu_0\}$ on the half-plane $\beta \leq 0$ (here $\mu_0 \simeq -0.74$). Inversion of the dependences $\alpha = A(\lambda, \mu), \beta = B(\lambda, \mu)$ and further calculation of quadratures with respect to α lead to a solution of the initial problem (22)–(25). Here the one-sided inequality is satisfied automatically.

Let us note some properties of this solution. The functions P and Q are continuous at the contact point $\alpha = \beta = 0$. At the same time, the functions

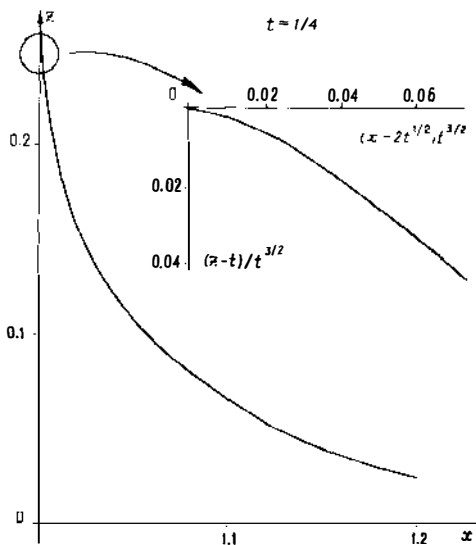


Figure 6 A plot of the free surface in initial and “stretched” coordinates.

Q_z and Π , which are proportional to the velocity tangent and pressure, respectively, have a discontinuity of the first kind at this point when passing from the wetted part of the contour to the free surface. (We recall that in the Wagner solution, the pressure possesses an integrable singularity at the contact point.) The pressure is maximal at the contact point; in dimensional variables we have $p_{\max} = \Pi_{\max} \rho V (kt)^{-1}$, where $\Pi_{\max} \simeq 0.47$, while at a distance of order $t^{3/2}$ from the contact point the pressure is of order $t^{-1/2}$ as $t \rightarrow 0$. Finally, the free boundary in the constructed solution, in contrast to that in the Wagner solution, approaches the contact point smoothly. (A free-boundary plot in “stretched” coordinates is presented in Figure 6.)

The Pointed Body

Wagner (1932) was first to study the problem of a wedge entering an ideal liquid that initially occupies a lower half-plane and is at rest. If liquid weight and compressibility are neglected, the initial problem possesses a self-similar solution depending on the two variables x/Vt , z/Vt , where V is the submergence velocity. As in classical jet theory, substantial progress can be achieved in the wedge-entry problem on the basis of methods using complex-variable function theory.

The main role in the analysis of the wedge problem is played by the so-

called Wagner function

$$h(\omega) = \int_{\infty}^{\omega} [d^2 W / d\omega^2]^{1/2} d\omega,$$

where $W(\omega)$ is the complex velocity potential, and $\omega = (Vt)^{-1}(x + iz)$. It is remarkable that the flow region in the plane of the Wagner function represents a closed triangle. This follows from the constant-pressure condition on a free boundary and the impermeability of the wedge face and axis of symmetry. Using the properties of the Wagner function and the methods of jet theory, Dobrovol'skaya (1965) reduced the problem to a single nonlinear integral equation for the real function $f(\tau)$ defined on the half-axis $\tau \geq 0$:

$$f(\tau) = C \int_{\tau}^{\infty} \frac{t^{-1-\alpha/\pi}(t+1)^{-1+\alpha/\pi} \exp\left(-\frac{1}{\pi} \int_0^{\infty} \frac{f(v)}{v-t} dv\right)}{\int_0^t s^{(-1/2)+(\alpha/\pi)}(s+1)^{-\alpha/\pi} \exp\left(\frac{1}{\pi} \int_0^{\infty} \frac{f(v)}{v-s} dv\right) ds} dt,$$

where 2α is the wedge opening angle, and C is a known functional of f . It should be noted that $f(\tau) = 0$ corresponds to the case when the free-surface deformation is negligible. The numerical analysis of this integral equation has been given by Dobrovol'skaya (1966, 1969). When the function $f(\tau)$ is known, all the process characteristics can be determined. The distribution of pressure along the wedge face at different values of α is shown in Figure 7.

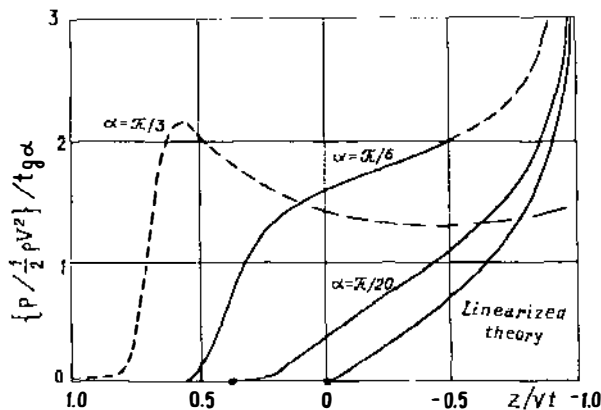


Figure 7 Pressure distribution along the wedge face for different values of the half-opening angle α (Dobrovol'skaya 1969).

A remarkable method of analyzing self-similar problems was developed by Hughes (1972). According to this method, a simple solution structure is specified near the singular points of the flow region corresponding to the physical nature of the phenomenon. In the rest of the flow region, a numerical calculation is carried out, which is improved by iterations with the account of the singularities available. This approach enables an adequate and economic algorithm to be constructed, which Hughes used to study the motion trajectory of liquid particles and the flow pattern at the apex of the splash jet. He (1972) notes, as does Dobrovol'skaya (1969), the nonmonotonic character of the pressure distribution along the wedge face.

Both the Dobrovol'skaya and Hughes approaches use the Eulerian flow description. Lagrangian variables were first used in such problems by Johnstone & Mackie (1973), who considered the plane problem of symmetrical wedge entry under the assumption that the particles on the free surface do not penetrate into the contact region at $t > 0$. In the plane of Lagrangian coordinates ξ, ζ , the flow region is the lower half-plane $\zeta < 0$, with a cut along the line segment $\xi = 0, -Vt < \zeta < 0$. The edges of this cut are preimages of the contact region. This approach yields a functional relationship between the half-opening angle α and the angle β between the tangent to the free boundary at the contact point and the wedge edge:

$$\operatorname{ctn} \beta + 2\beta = \pi + 2\alpha.$$

It should be noted that as $\alpha \rightarrow 90^\circ$, $\beta \rightarrow 9.5^\circ$ is obtained from this formula, whereas $\beta \rightarrow 0^\circ$ from the results of Dobrovol'skaya (1969). Such disagreement in the results is, the authors believe, due to nonuniqueness of the solution to the problem.

The construction and analysis of approximate models, for which the solutions have some peculiarities characteristic of more complicated problems of this kind, are of great importance for understanding the process dynamics. The problem of a thin body penetrating a liquid is one such model. It is natural to suppose that the disturbances due to thin-body penetration into liquid are small. Thus, the problem may be linearized about an initial state of rest, and its approximate solution obtained at finite values of time. The first results for this approximation were obtained for a compressible fluid by Grigoryan (1959) (axisymmetric case), and for an incompressible liquid by Mackie (1962) (plane and axisymmetric cases). The general solution constructed by Grigoryan (1959) allows penetration problems to be solved for arbitrary unsteady body motion. A generalization of the results obtained by Mackie (1962) for the inclined entry of a plane contour has been performed by Yim (1973) and Terentiev (1977).

It is noteworthy that in the problem of thin-wedge entry, the above approximation gives the wrong results near the tip of the wedge, where the velocity field possesses a logarithmic singularity. As Gonor (1986) showed, the dimensions of this region are of order $\exp(-1/\alpha)$ as $\alpha \rightarrow 0$. To describe the flow pattern more exactly, Gonor suggested using the problem of ideal liquid flow around the wedge. A continuous combination of this solution with the thin-body approximation permits us to construct (away from the contact points) uniformly valid asymptotics of the velocity field and pressure distribution as $\alpha \rightarrow 0$.

Non-self-similar penetration problems for pointed bodies have been considered by Yim (1985). The non-self-similarity is caused by both boundary curvature and the weight of the liquid. When choosing the calculation scheme, Yim notes the importance of the solution behavior near the splash-jet apex. Indeed, if the grid step near the jet apex is reduced, the calculation procedure becomes unstable. To overcome this shortcoming, it is necessary to introduce additional assumptions concerning the flow structure in a small vicinity of the three-phase contact curve. In this case there are different numerical solutions depending on the way that the splash-jet apex position and the velocity potential in its vicinity are defined; these should be chosen according to physical considerations (Yim 1985). Yim notes that for the convex pointed body, the numerical solution gives negative pressure in the vicinity of the splash-jet apex. This indicates separation of the splash jet from the body surface. For the concave wedge, the pressure along the solid surface is always positive.

The flow produced by a cone submerging along its axis of symmetry according to a power law is self-similar, as in the case of a wedge. However, the available results on this problem are few. For a blunt cone an asymptotic solution in the framework of the Wagner method has been given by Logvinovich (1969). Thus, in the case of symmetric, constant- V submergence of a cone with opening angle 2α , the drag force acting on the body from liquid of density ρ is determined by the formula

$$F = 4\rho V^2(Vt)^2(4\pi^{-1} \tan \alpha)^3[1 + o(1)] \quad \text{as} \quad \alpha \rightarrow \pi/2.$$

For a thin cone the corresponding formula, obtained by Sagomonyan (1974), is

$$F = \pi\rho V^2(Vt)^2\alpha^4 \ln(2\alpha)^{-1}[1 + o(1)] \quad \text{as} \quad \alpha \rightarrow 0.$$

The work of Popov (1979), who considers the linearized problem for finite opening angles of the cone, should also be mentioned. The solution of the problem is written in quadratures, and numerical results are obtained. Comparison with the known results for blunt and thin cones shows good agreement.

Bodies With Flattening

In this section we consider the penetration of a body with a plane front section into an ideal liquid for the case when the initial body contact with the liquid surface occurs over a finite area. The latter assumption enables one to apply the known solution for the impact of a disk floating on a liquid surface. The problem of floating-body impact was first formulated and solved by Joukowski (1884) for the case of impact between two spheres, one of which is half submerged in the liquid. The impact theory took its classic form in the works by Sedov (1934, 1939). In this theory, motions with sharp changes in the velocity of system points are considered, and by impact is meant the limiting regime in which the changes in velocity are discontinuous.

Let the floating-body velocity be zero when $t = 0$; it changes smoothly from zero to some constant value V_0 during $0 < t < \Delta t$ and is equal to V_0 for $t \geq \Delta t$. The main relations of impact theory then follow from the Euler equation and the corresponding boundary conditions after the successive limits $\Delta t \rightarrow 0$, $t \rightarrow 0$. The problem of floating-body impact reduces to a linear boundary-value problem for the velocity potential φ when the value $\partial\varphi/\partial n$ is given on the solid wall and $\varphi = 0$ on the free boundary, the displacement of flow-region boundaries being negligible during the time of impact.

It should be noted that the general penetration problem is an unsteady one, but that the sharp character of the velocity changes of liquid particles at the moment of impact allows the liquid motion to be defined immediately after initial contact using the solution of a simpler steady problem. The fluid motion after impact is defined by the body motion, and the whole effect reduces to the main momentum vector and to the main angular momentum vector, which are expressed through the additional masses and inertia moments (Sedov 1934). Their values depend on the body shape and dimensions and on the form of the domain occupied by the liquid; they are proportional to the liquid density and are independent of the body velocity.

The solution of the impact problem is greatly simplified if the body floats on a liquid surface occupying a lower half-space. In this case the problem can be reduced to that of the motion of a body in an infinite liquid. The impact of a plate of width $2b$ on incompressible liquid of finite depth h was investigated by Keldysh (1935). He showed that for $h/2b > 2.5$, the influence of the bottom is already insignificant. The corresponding problem for a disk (noncentral impact) was examined by Vorovich & Yudovich (1957). Its formulation is distinguished from the classical one in that the force acting on the disk and its point of application are given,

and the fluid motion and the law of motion of the disk are to be determined. The constructed solution makes physical sense if at every point on the disk, the impulsive pressure of the liquid is positive. As $h \rightarrow \infty$, only impacts at points in the circle of radius $b/5$ and concentric with that of the disk (where b is the disk radius) do not lead to separation. In the case of finite depth this circlediminishes, but for $h > b$ the decrease is insignificant.

The velocity field given by impact theory has an integrable singularity $|\mathbf{v}| \sim ar^{-1/2}$ in the vicinity of the contact line of the body with the liquid free surface. (Here r is the distance from the contact point, and a is a constant.) To elucidate the flow structure in this vicinity, Yakimov (1973) used local coordinates $\tilde{x}$, $\tilde{z}$ in this narrow zone. The "inner" problem obtained is a self-similar one with variables $\tilde{x}(t/a)^{2/3}$, $\tilde{z}(t/a)^{2/3}$. Its qualitative analysis permits one to describe the splash phenomenon, the closing of the cavity upon entry into the water of a body of finite dimensions, and a paradoxical degree of influence on the flow character in splash jets of a small parameter—the ratio of the densities of the atmosphere and the water.

A scheme to approximate the unsteady penetration process was suggested by von Kármán & Wattendorf (1929). Their method is a quasi-steady one, based on the assumption that at each moment the velocity field of liquid particles coincides with that immediately after impact on the floating body, for which the position at this moment is known. (The displacement of the free boundaries is ignored.)

To estimate the hydrodynamic loads acting at impact on a body with a plane front section, the compressibility of the fluid must be taken into account. As Logvinovich & Yakimov (1973) note, the results of impact theory should be considered as the limiting ones realized after the acoustic wave has spread into the liquid.

IMPACT OF A SOLID BODY WITH AN IDEAL COMPRESSIBLE FLUID

The theory of a body interacting with an ideal compressible fluid at a free surface has considerably progressed through studies of the dynamics of a drop impacting with a solid surface. The results of these studies related to the initial-impact stage and can be implemented with corresponding changes to more general problems. In the last few years, these results have been reviewed by Gonor & Rivkind (1982) and Lesser & Field (1983). Thus, here we focus our attention on the changes in this problem necessitated by liquid compressibility.

Denote by c_0 the sound velocity in the fluid at rest and by $M = V/c_0$ the Mach number, where V is the impact velocity. Then the solution of the

problem in the framework of an ideal incompressible liquid can be considered as the leading asymptotic term in the solution of the problem for a compressible fluid as $M \rightarrow 0$. In this limit the influence of compressibility on the flow structure is assumed small in comparison with the inertial liquid properties. But for the initial liquid stage, this hypothesis is invalid. Skalak & Feit (1966) and Logvinovich & Yakimov (1973) noted that in the case of a body with a plane front section penetrating an ideal, incompressible liquid the balance of mechanical energy is violated during an initial stage. For a blunt body the pressure value on the contact spot is of order $t^{-1/2}$ as $t \rightarrow 0$. Thus, in order to obtain real results for this stage of the calculation, the elastic properties of the fluid must be taken into account, whatever the Mach number.

Let us distinguish two characteristic cases of the initial stage of the process. In the first, there is a moment $t_* > 0$ such that for $0 < t < t_*$ the contact spot expands with a velocity exceeding the local sound velocity in the fluid. The disturbance front is attached obliquely to the contact line, and the disturbed fluid is bounded by a solid-body surface on one side and by the shock-wave front on the other. If a subsonic stage ($t_* < \infty$) follows the supersonic one ($0 < t < t_*$), it is expected for $t \gg t_*$, when the shock wave is far enough away from the contact line, the solution of the problem with compressibility neglected will describe the process well enough. In the second case, the supersonic stage is absent ($t_* = 0$), and the free-boundary deformation begins immediately after the moment of contact.

During the supersonic stage of the process, the flow pattern is described by a solution to the more general problem of the motion in an infinite compressible medium of a solid body whose volume changes according to a given law. It enables one to apply known approximate methods and exact gas-dynamics solutions. One of the main gas-dynamic approximations is the acoustic one, based on the assumption that the disturbance of the fluid is sufficiently small. The motion of the medium in this approximation is described by a velocity potential that satisfies the wave equation inside the domain; on the free boundaries it is equal to zero, and on the solid walls its normal derivative is given. According to this approximation, the submergence of thin bodies and the initial stage of impact with the blunt bodies at $M \ll 1$ may be considered.

A broad class of problems in which planar thin bodies and bodies of revolution with an arbitrary meridional section penetrate a compressible liquid was examined by Grigoryan (1959, and references therein). The description of the initial stage of impact with a blunt body is a more complicated problem because the contact-line position is unknown beforehand and is to be determined during solution. During the supersonic stage ($0 < t < t_*$), when the contact region is known, the solution can be written

in quadratures. Thus, the pressure distribution in the contact region for the supersonic stage of submergence of a plane, smooth contour as $M \rightarrow 0$ has been constructed by Rochester (1979) and Korobkin (1984) as follows:

$$p(x, t) = 2^{1/2} \pi \rho V c_0 \zeta^{-1/2} K \{ 2^{-1/2} [1 - (0.5 - \tau) \zeta^{-1}]^{1/2} \},$$

$$\zeta = [(\tau + 0.5)^2 - \alpha^2]^{1/2}, \quad \tau = c_0 t (MR)^{-1}, \quad \alpha = x (MR)^{-1}$$

$$\{ |x| < (2RVt)^{1/2}, 0 < t < t_* = M^2 R (2V)^{-1} \}.$$

Here $K(k)$ is the complete elliptic integral of the first kind, and R is the radius of the contour curvature at the initial contour point. From these formulae it follows that at $t = 0$, the pressure p equals the hydraulic impact pressure $\rho V c_0$, i.e. the pressure on a compressible half-space boundary if it is immediately accelerated to normal velocity V . Historically, this is the first formula to be used for calculating the strength of structures that during their use undergo impact with liquid masses. It was obtained by Cook (1928). For $t > 0$ the pressure decreases at the initial contact point, and its maximum value at fixed t is at the contact points, with the pressure at the contact point rising as t increases and becoming infinite at the moment that the disturbance wave emerges on the free boundary ($\tau = 1/2$).

During the subsonic stage ($t > t_*$) of impact, the flow can only be described numerically, analytical results being unknown. The position of the contact points is determined from the integral equation of Wagner (1932). This equation can be effectively solved for an incompressible liquid. At present no acoustic approximation with the analysis of the Wagner equation is known. Detailed numerical investigation of the plane problem of blunt-body entry into a weakly compressible fluid in the acoustic approximation has been carried out by Kubenko (1981a,b), Gavrilenko (1984), and Gavrilenko & Kubenko (1985). The latter authors propose a method of determining the position of the contact points for $t > t_*$ that is not connected with the integral-equation solution. An axisymmetrical problem for $0 < t < t_*$ has been considered by Shamgunov (1966).

The acoustic approximation obviously is invalid within a small vicinity of the contact points as $t \rightarrow t_* - 0$. The value of the flow disturbance in the fluid remains small as $M \rightarrow 0$, but the derivatives of the required functions in that zone are no longer small. A boundary-value problem describing plane fluid flow in the vicinity of the contact points as $t \rightarrow t_*$, $M \rightarrow 0$ has been formulated by Korobkin & Pukhnachov (1985). The equation for the velocity potential coincides with the nonlinear equation of unsteady transonic motion. The front position of the shock wave is unknown and is to be determined during the solution of the problem. The absence of initial conditions is a peculiarity of such an unsteady problem. These

conditions are replaced by the nontrivial condition after the limits

$$(t_* - t)M^{-8/3} \rightarrow \infty, \quad M \rightarrow 0, \quad t \rightarrow t_* - 0.$$

During the supersonic stage, the contact line belongs both to the shock wave and to the solid-body surface, which explains why the impermeability condition and the discontinuity conditions must be fulfilled on it. These relations form a closed system from which are determined both the pressure on the contact line up to the moment t_* at which the shock wave emerges on the free surface and the dependence of t_* on the parameters of the problem. This system was used by Heymann (1969) to construct an approximate solution to the problem in the vicinity of the contact line. Grigoryan (1978) investigated in detail the case of impact between two liquid masses. The corresponding relations in the problem of impact between a liquid drop and an elastic target were obtained and examined as $M \rightarrow 0$ by Lesser (1981). The problem of the submergence of a parabolic contour into an ideal compressible liquid, whose equation of state is taken in the Tait form $\rho/\rho_0 = (1 + np/\rho_0 c_0^2)^{1/n}$ (where ρ_0 is the liquid density at rest and $n \simeq 7.15$ for water) was considered by Korobkin & Pukhnachov (1985). The dimensionless pressure $p_* = p/\rho_0 V^2$ at the moment t_* at the contact points is defined by the following equation:

$$p_*[1 - \rho_0/\rho(p_*)] - 2 = M^2 p_*^2 [\rho_0/\rho(p_*)]^{n+1}.$$

Asymptotically as $M \rightarrow 0$, p_* has the form

$$p_* = [4/(n+1)]^{1/3} M^{-4/3} [1 + O(M^{1/3})].$$

The results obtained allow one to investigate the dependence of characteristic values on the Mach number. Thus, the pressure on the contact line is a nonmonotonic function of M (Grigoryan 1978), and its ratio to the hydraulic impact pressure $\rho V c_0$ is of order $M^{-1/3}$ as $M \rightarrow 0$, $t \rightarrow t_* - 0$.

When the supersonic stage is absent ($t_* = 0$), the problem of wedge submergence is a typical one. This problem has a self-similar solution constructed in the acoustic approximation by Sagomonyan (1974). For symmetric submergence with $M \tan \alpha < 1$, the drag force is given by

$$F = \pi M_0 [E(1 - M_0^2)]^{-1} \rho c_0 V^2 t \cdot \tan \alpha,$$

where $E(k)$ is the complete elliptic integral of the second kind with modulus $k = 1 - M_0^2$ ($M_0 = M \tan \alpha$). It should be noted that the integrated force of resistance to wedge entry is less than the hydraulic impact force of a plate of the same width at $t = 0$. Galin (1947) was the first to solve the problem of impact with a plate floating on a compressible liquid surface. The force $F(t)$ acting on a plate of unit length for $c_0 t \leq 2b$ (where $2b$

is the plate width) is given by

$$F = 2\rho c_0 Vb(1 - c_0 t/2b).$$

The formulation of the impact problem in the acoustic approximation is still rather complicated. A simplification based on the hypotheses of geometric acoustics was presented by Lesser (1981). In this model a definition of the pressure distribution behind the disturbance-wave front during the supersonic stage is achieved with an arbitrary geometry of the flow-region boundaries. Comparison of the analytical formulae obtained with an exact solution suggested by Rochester (1979) in a plane case and with numerical calculations by Rosenblatt et al. (1979a,b) in a three-dimensional case yields good agreement.

It should be noted that during construction of approximate impact models, a common assumption is that rarefaction zones in the flow region are absent. However, this assumption runs counter to both the well-known experimental data obtained by Brunton & Camus (1970) and the numerical calculations by Gonor & Yakovlev (1977, 1978). The analysis becomes more complicated because of the cavitation processes taking place during impact between the solid and liquid masses. That is why constructing averaged models on the basis of an integral approach seems an attractive prospect. Experimental data can serve as a basis for rational simplification of the problem. Thus, Eroshin et al. (1980) showed that unlike the situation for bodies with flattening (plate) and pointed bodies (cone), the drag force for a blunt body (hemisphere) only depends weakly on the Mach number when $0 < M < 0.75$ (Figure 8). It is therefore expected that for blunt bodies, the incompressible-liquid model gives a correct idea of the global properties of the process at small Mach numbers.

There has been much experimental and numerical investigation of impact between a solid body and a compressible liquid, but as yet there is no well-developed theory leading to an unambiguous interpretation of

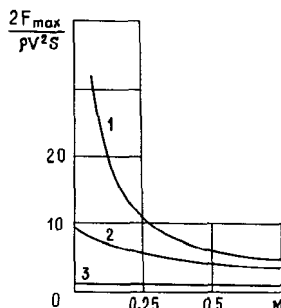


Figure 8 The maximum values of the drag force for a disk (1), a cone with half-opening angle $\alpha = 75^\circ$ (2), and a hemisphere (3) versus the Mach number (Eroshin et al. 1980). S , midsection area.

the results. Even within the framework of simplified models, the flow pattern at the moment of shock-wave emergence at the liquid free boundary is not clear. There are some reasons to believe that it is at this stage that the impulse thin-jet ejection takes place, as has been noted in experimental work by Stebnovsky (1979) and Stow & Hadfield (1981).

As a blunt body submerges into a slightly compressible fluid ($M \ll 1$), the duration of the stage during which the elastic fluid properties are important is several times greater than $t_* = RV/2c_0^2$. Under real conditions, t_* is of order several microseconds. (For $R = 1$ m, $V = 100$ m s⁻¹, $c_0 = 1500$ m s⁻¹, we have $t_* = 2.2 \times 10^{-5}$ s.) Thus, application of the incompressible-liquid model for $M \ll 1$ and $t_*/t \ll 1$ is justified. It is thought that the problem of blunt-body submergence can be correctly formulated within the framework of this model under the condition of flow kinetic energy finiteness, i.e. additional information about the liquid properties is not necessary. However, taking compressibility into account, it is possible to define the fine flow structure at the initial stage. On the other hand, at time t such that $t_* \ll t \ll R/V$, the characteristics of the process can be calculated by the method described in the previous section.

COMPLICATED IMPACT MODELS

In some cases, an adequate description of the impact process between solid and liquid masses requires that a more complicated model be used. For example, in ship building it is essential to take into account the elastic hull properties, when calculating the hull strength. Similarly, in order to understand fully the mechanism of solid-body erosion at the point of impact of liquid drops with the solid-body surface, one must consider the strength properties of bodies.

Complete reviews of the main achievements in studying the problem of impact between a liquid and a deformable body have been presented by Grigolyuk & Gorshkov (1976), Gorshkov (1981), and Kubenko (1981a). Most results were obtained numerically because of the necessity to define both the liquid motion and the strained-state structure.

Sharov (1958) was apparently the first to consider the joint initial problem of impact of a thin-walled structure with a liquid as a sequence of two problems. First, the liquid flow and the loads acting on the submerging body are determined without taking into account elastic deformation. Second, the tangential and normal W components of the elastic displacements are calculated according to the known load distribution. The asymptotic character of this assumption as $t \rightarrow 0$ has been demonstrated by Korobkin (1983) and Korobkin & Pukhnachov (1985). Solutions to plane and axisymmetrical problems are written in quadratures. In addition, the

character of the deformation at small t is defined by a single essential parameter

$$q = 3VR[12(1-\nu^2)\rho_1/E]^{1/2}/4h,$$

where ρ_1 is the density of the material shell, h its thickness, E the modulus of elasticity of the material, ν its Poisson's ratio, and R the curvature radius at the shell apex. The expression for the deflection of a shell of revolution at its top is given by

$$W_0(t) = 4(R/3)^{1/2}\rho(\pi h\rho_1)^{-1}q(\text{ci } q \sin q - \text{si } q \cos q)(Vt)^{3/2}[1 + O(t^{1/2})]$$

as $\bar{t} \rightarrow 0$,

where $\text{ci } q$ and $\text{si } q$ are the cosine and sine integrals.

In our discussion of impact between liquid and solid masses, we have ignored the influence of the medium in which these masses are found. The model of incompressible liquid leads to infinite pressure, and the model of compressible liquid to a pressure jump, at the moment of contact. When the surrounding medium is considered, the free boundary of the fluid is disturbed before the body touches it. Two cases are possible. In the first, owing to the liquid free-boundary deformation, a cavity is formed that is closed at the moment of contact. In the second case, the cavity is absent because the curvature of the deformed free boundary is small, and the liquid surface at the moment of contact has a velocity equal to that of the body. In addition, the pressure affecting the body from the air and then from the liquid has no singularities at $t = 0$.

Theoretical and experimental studies of the problem of a flat plate dropped upon a liquid surface suggest that the first case describes this process. Important work on this problem has been done by Verhagen (1967), Asryan (1972), Koehler & Kettleborough (1977), and Eroshin et al. (1984). The second case, as Moran & Kerney (1965) have shown, takes place if a stretched "cigar-shaped" body approaches the free surface.

The stage considered is characterized by the presence of a so-called air-cushion between the solid and liquid masses. The problem of describing the system at this stage belongs to a wide class of problems concerning solid-body motion near the interface between media. However, the air-cushion problem has some peculiarities. For example, attention is focused on the system behavior at the moment of contact, whereas in the classical theory the possibility of contact is not considered. As a consequence, there is no satisfactory understanding of the air cushion damping mechanism. Thus, the main question in mathematical modeling of cavity formation at the moment of contact is solved a priori.

Until now we have discussed only the initial stage of the process in which a body with constant velocity enters into an incompressible liquid occupying a half-space. When the submergence velocity $V(t)$ is variable, the problem becomes more complicated. Approximate approaches to its solution based on the integral conservation laws have been presented by Logvinovich (1969). It should be noted that in one special case, when $V = Ct^n + O(t^{n+1})$ as $t \rightarrow 0$ (C and n being positive constants), this problem possesses an asymptotic self-similarity property. The leading asymptotic term in this case can be calculated in terms of the solution of a mixed problem for a harmonic function (Pukhnachov & Korobkin 1981).

If a body is submerged into a basin of finite dimensions, then even with constant velocity the leading term of the outer asymptotic expansion as $t \rightarrow 0$ loses its self-similarity, which makes investigation of the process more difficult. Numerical methods for solving the linearized form of the problem have been developed by Galanin & Saikin (1981, 1983). Another approach, based on the theory of variational inequalities, has been proposed by Korobkin (1982).

The previous considerations were based on the intuitively evident assumption that at the initial impact stage, the liquid inertia forces dominate the forces of surface tension, gravity, and viscous interaction. Indeed, let us estimate the characteristic values of the Weber number $We = \rho V^2 R / \sigma$, the Froude number $Fr = V^2 / gR$, and the Reynolds number $Re = VR / \nu$, where σ is the surface-tension coefficient, g the acceleration due to gravity, and ν the kinematic viscosity coefficient. With $R = 10$ cm, $V = 10^4$ cm s⁻¹, $\rho = 1$ g cm⁻³, $\sigma = 72.5$ dyne cm⁻¹, $g = 981$ cm s⁻², and $\nu = 0.01$ cm² s⁻¹, we obtain $We = 1.38 \times 10^7$, $Fr = 1.02 \times 10^4$, and $Re = 10^7$. To make these ideas more convincing, we note that capillarity affects the second term, and gravity only the third term, of the outer asymptotic expansion as $t \rightarrow 0$. (The corrections to the displacement vector of the liquid particles, $\mathbf{X}$, necessitated by these factors are of order $t^{3/2}$ and t^2 , respectively.) When taking gravity into account, it is not necessary to modify the leading term of the inner expansion. The contribution of capillary forces is comparable with that of inertia forces in a vicinity of the contact point with radius of order $We^{-1}Vt$. The time interval over which this contribution is to be taken into account is estimated by $t_0 = RV^{-1}We^{-2}$, which is negligible under the conditions considered.

It is more difficult to present a quantitative estimate of the role of the viscosity at the initial impact stage. From general arguments it is expected that for $Re \gg 1$ the main viscous action is localized in the boundary layer near the contact spot having thickness of order $(\nu t)^{1/2}$ as $t \rightarrow 0$. Let us evaluate approximately the contribution of viscosity to the drag force of a parabolic contour having a curvature radius R at the apex when entering

an incompressible liquid at constant velocity V . If we ignore the contribution from small vicinities of the contact points when calculating the unsteady boundary layer on the wetted part of the contour, the method of Stewartson (1960) can be used. It leads to a viscous component of the drag force $O[\rho V^2(\nu t)^{1/2}]$ as $t \rightarrow 0$. At the same time, the value of the drag force for the contour entering an ideal liquid is given for small t by the expression $2\pi\rho V^2 R[1 + O(t^{1/2})]$. The ratio of these two values stays small so long as the thickness of the unsteady boundary layer is small compared with the characteristic linear dimension R .

The description of the viscous flow near the contact points or the contact line is inseparable from the solution of the known problem of the dynamic edge angle. Although some progress has been made on this problem, (Dussan V. 1979, Cox 1986), practically all its considerations concern the case of a constant-velocity three-phase contact line. Such a situation, or a similar one, is characteristic of a wedge or pointed body submerged with a constant velocity V , but for a blunt body the contact-point velocity is of order $(RV/t)^{1/2}$ as $t \rightarrow 0$. In the latter case it is natural to accept the hypothesis that the liquid rolls on a solid wall like an unrolling carpet and that the dynamic edge angle takes the value π (Pukhnachov & Solonnikov 1982). This hypothesis leads to a closure of the problem formulation under the framework of the Navier-Stokes equations, with the natural conditions on the free boundary and the no-slip conditions on the solid wall. Thus, it is hoped that there will be agreement between the viscous-flow asymptotics near the contact line and the inviscid limit leading to a dynamic edge angle value asymptotically equal to π (Korobkin & Pukhnachov 1985).

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